

1. **Team members:** Bryce Clement, Christian Rutherford, Devon Diaco
2. **Project Title (tentative):** “Predictive Weight Optimisation: A Machine Learning Approach to Maximising Expected Value in Men’s March Madness Brackets”
3. **Phenomenon to be modelled:** The phenomenon being modelled is the outcome of the 2026 NCAA Division 1 Men’s Basketball tournament, otherwise called “March Madness.” This is a single-elimination tournament known for its high variance, “upsets” being frequent, and non-linear relationships between victories and the past records of a team.
4. **Goal:** Our project’s objective is to develop a machine learning model determining optimal weights for various performance metrics to maximise the expected value of a bracket. Instead of simply predicting individual game winners, we will be attempting to optimise these weights in order to produce a bracket maximising the total point yield based on the ESPN Tournament Challenge scoring system (in which later rounds are weighted more heavily).
5. **Modelling Methods:** To achieve our objective, we will be applying various methods typical to the machine learning model process:
 - a. **Feature engineering:** we’ll be aggregating historical tournament data from the previous 10-15 years, normalising stats such as Effective Field Goal Percentage, strength of schedule, and the adjusted efficiency margin. We will also implement a momentum feature, applying a decay weight to late-season team performance & conference tournament results in order to capture team trajectory heading into March.
 - b. **Logistic Regression and Random Forest (tentative, though likely):** As a baseline, we’ll be applying logistic regression to determine the probability of a “win” for a given matchup. Random forest will assist us in identifying non-linear feature importance, such as whether a 3-point percentage matters more in the Sweet 16 than in the Round of 64.
 - c. **Gradient boosting (XGBoost/LightGBM/Catboost likely):** We’ll utilise gradient boosting to refine the weights of our predictive variables, with the intention of minimising their log loss.
 - d. **Expected Value Optimisation:** We will also incorporate public sentiment data (ESPN’s “who picked whom” statistics), by identifying leverage opportunities where our model’s win probability significantly exceeds public consensus, we can optimise for the EV of a bracket. This enables the team to identify high-upset potential necessary for 99th-percentile finishes.

- e. **Monte Carlo Simulation:** Following the establishment of our weights, we'll run thousands of tournament simulations to determine which "optimal" bracket configuration has the greatest frequency of finishing in the 99th percentile of ESPN pools.

6. General references:

- a. <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>
- b. https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html
- c. https://en.wikipedia.org/wiki/Gradient_boosting
- d. https://xgboost.readthedocs.io/en/release_3.2.0/
- e. <https://lightgbm.readthedocs.io/en/stable/>
- f. <https://catboost.ai/>
- g. <https://kenpom.com/>
- h. <https://www.ncaa.com/stats/basketball-men/d1>
- i. <https://arxiv.org/abs/1412.0248>
- j. Will be looking for various other papers/sources on applying the above models for our purposes

7. Changes to methods will be made as needed, in the event that problems arise that prevent us from achieving our goals